Complex Tori and Elliptic Curves

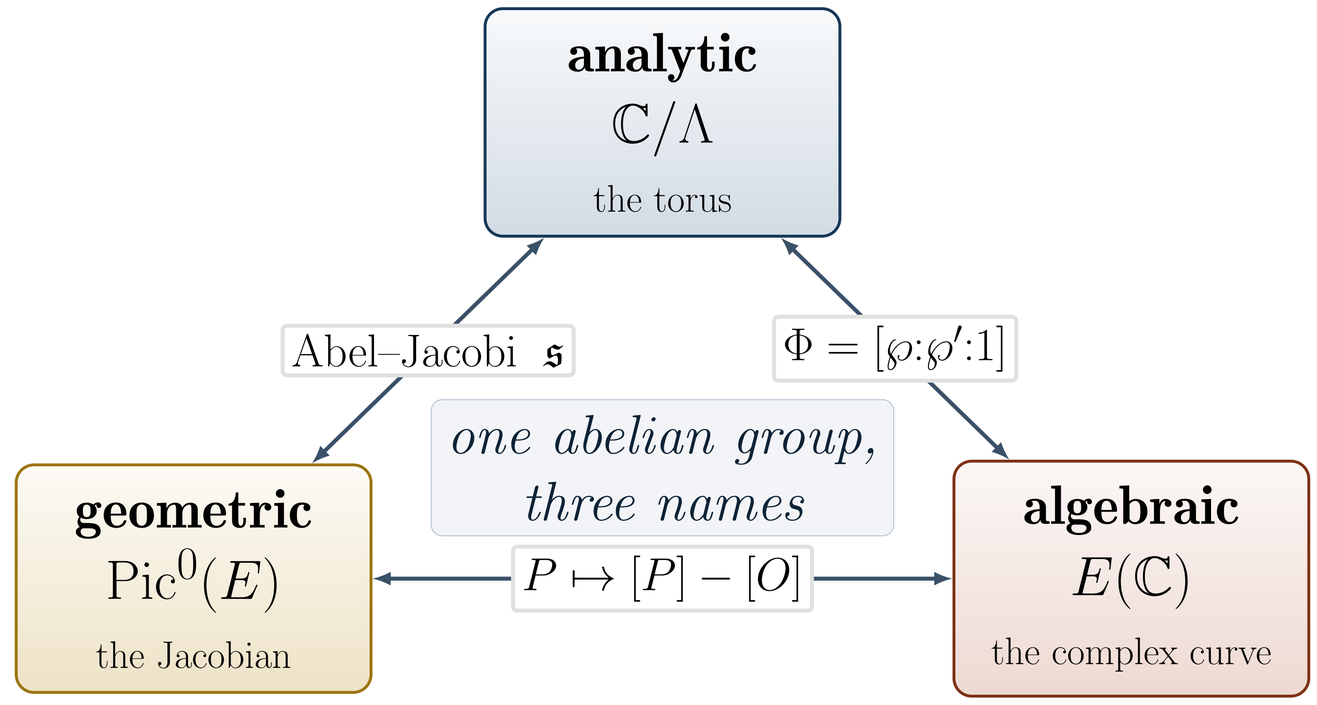
One Object, Three Faces — over $\mathbb{C}$

The same compact complex Lie group as a torus $\mathbb{C}/\Lambda$, a Jacobian $\mathrm{Pic}^0(E)$, and a curve $E(\mathbb{C})$

One Object, Three Faces, over $\mathbb{C}$

A single object — a compact one-dimensional complex Lie group, a *one-dimensional abelian variety* — appears in three guises: the torus $\mathbb{C}/\Lambda$ (analytic), the Jacobian $\mathrm{Pic}^0(E)$ (geometric), the projective curve $E(\mathbb{C})$ (algebraic). **Uniformization makes all three one group**, by explicit biholomorphisms. Columns 1–2 develop each face; column 3 specializes the field to $\mathbb{F}_p$ for the arithmetic frontier.

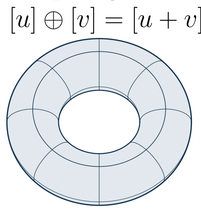
The thesis. $\mathbb{C}/\Lambda \cong \mathrm{Pic}^0(E) \cong E(\mathbb{C})$ as complex Lie groups; analysis, geometry, and algebra describe *one* object.



The maps: $\Phi = [\wp:\wp':1]$ (torus to cubic); $P \mapsto [P] - [O]$ (curve to Jacobian); Abel–Jacobi $\mathfrak{s}$ (divisor to a point of $\mathbb{C}/\Lambda$). Compose around the triangle and return to the start.

1. Analytic: the torus $\mathbb{C}/\Lambda$

A **lattice** $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ ($\omega_2/\omega_1 \in \mathbb{H}$) tiles $\mathbb{C}$; $T = \mathbb{C}/\Lambda$ is topologically $S^1 \times S^1$, a compact **genus-1 Riemann surface** and a **complex Lie group**, $[u] \oplus [v] = [u+v]$.



$\mathbb{C}$ tiled by Λ, glued into the torus — the group carried to all faces.

The Weierstrass function. The simplest non-constant elliptic function,

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda \neq 0} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right),$$

has one double pole per cell. With the Eisenstein series $g_2 = 60G_4$, $g_3 = 140G_6$ it obeys $\wp'^2 = 4\wp^3 - g_2\wp - g_3$, and $\mathbb{C}(\wp, \wp')$ is the *entire* field of elliptic functions.

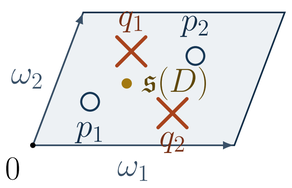
Periods & uniformization. The holomorphic 1-forms are exactly $\mathbb{C}\,dz$, with $\oint_{\gamma_i} dz = \omega_i$, so Λ is the period lattice. Conversely every genus-1 surface with a base point is such a $\mathbb{C}/\Lambda$, unique up to scaling.

2. Geometric: divisors, Riemann–Roch, Pic^0

A **divisor** $D = \sum n_i[P_i]$ has degree $\sum n_i$; principal divisors $\mathrm{div}(f)$ have degree 0.

Riemann–Roch. On genus 1, $\ell(D) = \deg D$ for $\deg D > 0$. Hence $\ell(nO) = n$, and $1, \wp, \wp'$ span $L(3O)$ — *these are the functions that embed E as a plane cubic.*

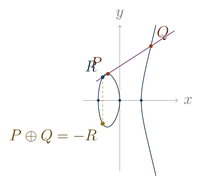
The Jacobian. $\mathrm{Pic}^0(E) = \mathrm{Div}^0(E)/\{\mathrm{div}(f)\}$, the degree-0 line bundles. By **Abel's theorem** (principal $\iff$ degree and sum vanish), the sum map $\mathfrak{s}(\sum n_i[P_i]) = \sum n_i z_i$ gives $\mathrm{Pic}^0(E) \cong \mathbb{C}/\Lambda$, and $P \mapsto [P]-[O]$ makes E its own Jacobian, $E \cong \mathrm{Jac}(E)$.



Abel–Jacobi sends a degree-0 divisor to $\sum n_i z_i$, vanishing exactly on principal divisors — so $\mathrm{Pic}^0(E) = \mathbb{C}/\Lambda$.

3. Algebraic: the curve $E(\mathbb{C})$

The embedding $\Phi = [\wp:\wp':1]$, $\Phi(0) = O = [0:1:0]$, is a **biholomorphism** of $\mathbb{C}/\Lambda$ onto the smooth cubic $E : y^2 = 4x^3 - g_2x - g_3 \subset \mathbb{CP}^2$ — a projective curve of genus 1 and a one-dimensional **abelian variety**.



Chord–tangent on E: the line through P, Q meets E again at R; reflecting gives $P \oplus Q$. Over $\mathbb{C}$ this is the transported torus addition.

An algebraic group. The law is *rational* in g_2, g_3 (Vieta), $x_{P\oplus Q} = \frac{m^2}{4} - x_P - x_Q$, so E is a **group scheme**. Its torsion is $E[2] = \{O, e_1, e_2, e_3\}$ (the roots) and $E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$.

Smooth vs. singular. E is smooth iff $\Delta = g_2^3 - 27g_3^2 \neq 0$; as $\Delta \to 0$ the torus pinches to a *node* (group $\mathbb{C}^*$) or a *cusp* ($\mathbb{C}$). The class is $j = 1728\, g_2^3/\Delta$.

3′. Cohomology and moduli (over $\mathbb{C}$)

The Jacobian is also Hodge data — the exponential sequence of sheaves realizes the same group:

$$0 \to \mathbb{Z} \xrightarrow{2\pi i} \mathcal{O}_E \xrightarrow{\exp} \mathcal{O}_E^* \to 0$$

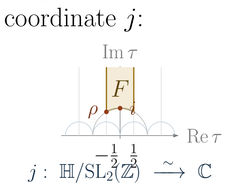
take cohomology

$$\mathrm{Pic}^0(E) = \frac{H^1(E, \mathcal{O}_E)}{H^1(E, \mathbb{Z})} = \frac{\mathbb{C}}{\Lambda}$$

$H^1(E, \mathcal{O}_E) \cong \mathbb{C}$ $\qquad$ $H^1(E, \mathbb{Z}) \cong \mathbb{Z}^2 = \Lambda$

$\mathrm{Pic}^0(E) = H^1(\mathcal{O}_E)/H^1(\mathbb{Z}) = \mathbb{C}/\Lambda$; *the Hodge piece $H^{1,0} = \mathbb{C}\,dz$ is Face 1's holomorphic form.*

All such objects are organized by their shape $\tau \in \mathbb{H}$ modulo $\mathrm{SL}_2(\mathbb{Z})$, with global coordinate j:



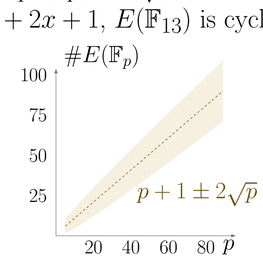
$\mathbb{H}/\mathrm{SL}_2(\mathbb{Z}) \xrightarrow{\sim} \mathbb{C}$ *via the j-invariant; i, ρ are the extra-symmetric curves.*

One object. Over $\mathbb{C}$, $\mathbb{C}/\Lambda$, $\mathrm{Pic}^0(E)$ and $E(\mathbb{C})$ are three *presentations* of one abelian variety — not merely isomorphic, but biholomorphic.

4. Specialize: over $\mathbb{F}_p$, the frontier

The group law is rational, so E makes sense over any field. Over a finite field the three faces collapse to a finite group — and a new world opens.

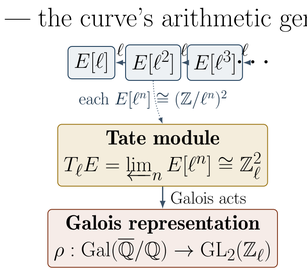
Finite fields & Hasse. $E(\mathbb{F}_p)$ is a finite abelian group with $\#E(\mathbb{F}_p) = p + 1 - a_p$, $|a_p| \leq 2\sqrt{p}$: a band of width $4\sqrt{p}$ about $p+1$. (For $y^2 = x^3 + 2x + 1$, $E(\mathbb{F}_{13})$ is cyclic of order 8.)



Real counts $\#E(\mathbb{F}_p)$ for $y^2 = x^3 - x + 1$, $p \leq 89$: all inside the Hasse band $p + 1 \pm 2\sqrt{p}$.

Cryptography. Adding is fast but *dividing* is not: recovering n from nP is the elliptic-curve discrete logarithm, the hardness behind ECC and ECDSA.

Galois and the Tate module. The torsion $E[\ell^n]$ carries a $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-action — the curve's arithmetic genome.



The torsion levels form an inverse system; its limit $T_\ell E \cong \mathbb{Z}_\ell^2$ carries $\rho : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{Z}_\ell)$.

Modularity, BSD, CM. Every $E/\mathbb{Q}$ is *modular* (Wiles — the theorem behind Fermat); the rank of $E(\mathbb{Q})$ is governed by Birch–Swinnerton-Dyer; and CM curves (like $y^2 = x^3 - x$, $j = 1728$) tie to class field theory.

One identity, two worlds. Over $\mathbb{C}$ the three faces are one geometry; over $\mathbb{F}_p$ or $\mathbb{Q}$ the same law is arithmetic. The dictionary runs in every direction.

Author: ⟨Your Name⟩ • ⟨email⟩ • seeking PhD positions in arithmetic & algebraic geometry.

Companion to the main poster and the gallery.

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